

Assignment 6

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data sent by Frank. We have a new transmission:

Mission Log 006 — from Frank

Hey hey hey, Analyst, it's me, got a weird one for you today. Command rolled out this fleet-wide Wellness Initiative a few weeks back. Every ship in the IRO fleet checks in crew wellness on a rotating basis, and it all funnels back through the *Meridian*'s comms array because apparently we're "unusually stable." Thanks, I did maintain it.

Before *Cinderspore*, IRO had years of baseline data on how crews typically report their wellness. Command wants to know if the current picture has actually drifted from that old baseline, or if it just looks that way. And I want to know something too: is it the same for every job on this ship, or are some roles getting hit harder than others? I have a theory.

Data's attached. It's bigger than our usual crew roster since it's fleet-wide, not just us 30, so don't panic when you see the row count. Go find out if I'm right.

Variable	Type	Description
crew_id	ID	Unique fleet-wide personnel ID
crew_role	Categorical (4 levels)	Command, Engineering, Science, or Medical
wellness_status	Categorical (4 levels)	Thriving, Steady, Strained, or Critical
containment_team	Categorical (2 levels)	"Yes" if assigned to the Containment Response Team, otherwise "No"
leave_requested	Categorical (2 levels)	"Yes" if the crew member requested leave this reporting period, otherwise "No"

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's data here.

Question 1: Has the Wellness Picture Drifted?

Frank: "IRO's had years of baseline data on this. Before all the Cinderspore chaos, wellness checks across the fleet have always landed close to: Thriving 30%, Steady 40%, Strained 20%, Critical 10%. Tell me if we still look like that, or if this mission changed us."

Question: Does the current fleet-wide distribution of wellness status (`wellness_status`) match the historical IRO baseline (Thriving 30%, Steady 40%, Strained 20%, Critical 10%)?

1a. Find the observed count and percentage of crew members in each `wellness_status` category.

1b. Perform the appropriate hypothesis test to determine if the current wellness distribution differs from the historical baseline.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

Question 2: Is Wellness Tied to Job Role?

Frank: "Here's the one I actually care about. I've got a theory that some jobs on this ship are just harder on people than others, wellness-wise. Tell me if I'm imagining it."

Question: Is wellness status (`wellness_status`) independent of crew role (`crew_role`), or does wellness status depend on which job someone has?

2a. Construct the contingency table of `wellness_status` by `crew_role`.

2b. Perform the appropriate hypothesis test to determine if wellness status is independent of crew role.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

Question 3: Is the Containment Team Requesting More Leave?

Frank: “Small team, big rumor. Fifteen people on Containment Response, and Command swears they’re putting in leave requests way more than everyone else. Find out if that’s real.”

Question: Is leave requested (`leave_requested`) associated with Containment Response Team membership (`containment_team`)?

3a. Construct the 2x2 contingency table of `leave_requested` by `containment_team`.

3b. Show that we do not meet the assumption for the standard chi-square test of independence by checking the expected cell count for `leave_requested` = “Yes” and `containment_team` = “Yes”. You must show your work and justify your answer.

3c. Keeping your answer to the previous question in mind, perform the appropriate hypothesis test to determine if leave requests are associated with Containment Response Team membership.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...